

Landing error scoring system: A scoping review about variants, reference values and differences according to sex and sport[☆]

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ABSTRACT

Objective: The Landing Error Scoring System (LESS) is a movement analysis tool proposed to identify the risk of anterior cruciate ligament injuries, very useful for injury prevention. The aim of this study is to review the variants of the LESS, their normative scores and the differences according to sex and sport practiced.

Methods: PubMed, Scopus and ScienceDirect databases were searched from inception to October 19, 2023. Studies were eligible if the objective was finding normative or reference scores for the LESS, analyze the differences between sexes or sports, or used some variant of the test. Results were limited to available full-text articles published in English in peer-reviewed journals.

Results: Of the 360 articles identified, 20 were included for a full analysis (18,093 participants, age = 8–30 years, males = 70.6%). The military population was the most frequently analyzed (7 studies, n = 16,603). Results showed six variants of the LESS and average values ranged from 2.56 to 7.1. Males and females showed different pattern landing with errors in different planes.

Conclusions: Our findings highlight the need for more field studies on LESS reference scores, particularly for females and basketball or hockey players. Further research is required before conducting a systematic review and meta-analysis.

1. Introduction

The tools used to evaluate sports movement quality are essential for assessing and monitoring athletes. Several tests have been developed with the aim of identifying deficits in neuromuscular ability that are associated with the risk of injury (Cook et al., 2014; Gribble et al., 2012; Plisky et al., 2021). The Landing Error Scoring System (LESS) is a movement analysis tool developed by Padua (Padua et al., 2009) in 2009 to identify the risk of sustaining non-contact anterior cruciate ligament (ACL) injuries and biomechanical movement patterns involved.

The LESS protocol involves both horizontal and vertical jumps. During this activity, the participant must jump out horizontally from a 30 cm high box or platform to a distance that is 50% of their height. The landing must be made in this predefined zone, immediately followed by a maximum vertical jump, landing again in the same designated area.

During the instructions, it is emphasized that the participant should strive to reach the highest possible height in the vertical jump after landing. To capture the performance of the participants, two video recording devices are used, positioned 3 m from the landing zone at a height of 1 m above the ground. It is recommended that cameras have a sampling rate higher than 60 Hz, with a value higher than 100 Hz being optimal. This configuration allows for a greater perception of each phase during the jump. The score obtained in the LESS is calculated by a simple observational evaluation of movement errors in the landing technique (Padua et al., 2009).

The original protocol consists of 17 items that are evaluated to determine the total score in the LESS. A higher score on the LESS indicates a worse jump landing technique. The test includes the evaluation of the lower extremity and trunk positioning at the time of moment of initial contact (IC), the moment of maximum flexion of the knee (MKF),

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and an assessment of global movement, among others. LESS is scored upon landing from the horizontal jump off the box. The purpose of the vertical jump is to add an additional task to "distract" the participant and provide a sport-specific context. Thus, a true assessment can be made without the participant solely focusing on good form during a single landing. In the literature, most studies have included three attempted jumps and extracted the final LESS score from either one of them, which could be the first jump (Onate et al., 2010), the third jump (O'Malley et al., 2017), the best of the three (Garbenytė-Apolinskienė et al., 2018), the average of the three jumps or the sum of errors. It should be noted that these different calculation methods will have an impact on the final LESS scores (Hanzlíková, Athens, & Hébert-Losier, 2020). In fact, not only the final LESS score but also the risk categorization could be different depending on the calculation method used. This highlights the heterogeneity of the study designs.

Previous reviews have examined the method's capability to detect changes in landing mechanics, its validity, factors that influence the results, and its reliability. Grade B evidence suggests that the LESS is an effective tool for detecting changes in landing mechanics following the implementation of a prevention program (Pointer et al., 2017). The validity against 3D motion capture for most of the items addressing key risk factors for anterior cruciate ligament (ACL) injury has been found to be moderate-to-excellent (Hanzlíková et al., 2020b). Also, it has been described certain factors, such as a history of anterior cruciate ligament injury or a program of neuromuscular training, may impact the biomechanics of the jump and this, in turn, could modify the results. Particularly, participants with previous anterior cruciate ligament injury have higher LESS scores than healthy controls whereas neuromuscular training programs lasting a minimum of six weeks and other intervention programs decrease LESS scores (Hanzlíková, Athens, & Hébert-Losier, 2021). Regarding its degree of agreement, at least a good interrater reliability (Markbreiter et al., 2015), as well as good-to-excellent intrarater and intersession reliability (Hanzlíková et al., 2020b), have been reported.

The capacity of injury prediction has been evaluated in some follow-up studies. The LESS has demonstrated a sensitivity of 86% and specificity of 64% to predict ACL injuries among elite youth soccer athletes (Padua et al., 2015). Similar values were obtained for musculoskeletal injuries in military recruits (Everard et al., 2018). However, another study did not find a significant relationship between the risk of suffering ACL injury and the LESS score in college and high school athletes. Nevertheless, the authors highlighted the need to include more participants in the study for optimal statistical power (Smith et al., 2012).

The aim of this review was to explore and summarize current literature about LESS variants, normative scores and differences according to sex and sport. As far as we know, this is the first review boarding these topics. It will help to identify knowledge gaps and determine whether a systematic review of the literature is needed (Munn et al., 2018). Also, it will be useful for athletic trainers and physical therapists for a better understanding of the expected values in their sports players. By elucidating these factors, practitioners can choose a specific variant of the test according to their needs, enhance injury prevention strategies, and optimize rehabilitation protocols tailored to individual athletic profiles.

2. Methods

2.1. Literature search

Based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) method (Tricco et al., 2018), we identified studies that focused on variants, normative scores, and differences according to sex and sport for the LESS test.

From the results we expected to identify the coverage and the volume of studies available focusing on those areas, and the main characteristics of the methodology used in the available literature. Unlike other types of

reviews, scoping reviews are useful to address topics where heterogeneous study designs might be applicable. Also, it was within the scope of the study to identify knowledge gaps and the possibility of conducting a systematic review in relation to the topics of interest. The purposes described above, fit within the exploratory purposes described as indicative of a scoping review (Munn et al., 2018, 2022; Peters et al., 2020; Tricco et al., 2018). Hence, a scoping review was deemed appropriate as the review method selected.

Our process involved several steps. First, we conducted a systematic search. A preliminary review of studies using the LESS revealed substantial heterogeneity in their designs and variability in the amount of information and level of details provided. Therefore, following reported methodological guidance (Peters et al., 2020), it was considered appropriate to include only traditional publishing channels that ensured a minimum of standardized information. Under this assumption, three databases were used for the systematic search and grey literature (i.e., information produced outside of traditional publishing channels) was excluded. Second, we removed duplicate entries. Third, we applied inclusion and exclusion criteria to screen the remaining articles based on their titles and abstracts. Fourth, data extraction was carried out. Finally, a narrative review was conducted, organized by topic.

Since this study is a scoping review, there was no requirement to register a protocol with the International Prospective Register of Systematic Reviews (PROSPERO).

PubMed, Scopus and ScienceDirect databases were searched from inception to October 19, 2023 with the following term: "landing error scoring system" (quotes included). A web-tool was used to export the results, and duplicates were subsequently eliminated. Two independent reviewers (MA and LD), who were blinded to each other's decisions, screened the titles and abstracts of the articles to apply the inclusion and exclusion criteria. In case of disagreements, a third reviewer (PC) made the final decision.

2.2. Study selection

The inclusion criteria required that the study reported original and peer reviewed data about normative or reference scores of LESS in healthy individuals, and/or differences in values according to sex or sport. The articles must have objectives and/or hypothesis referring to this data. Also, we included articles that used a variant of the test. Exclusion criteria were: studies published in languages other than English, partial use (or minimal use) of the LESS test, reviews, background articles, case reports and data reported in conference, books or abstracts. Differences between reviewers regarding selection eligibility were resolved by majority decision after a third reviewer (PC) considered the full-text records and applied the selection criteria. Study characteristics were independently extracted from each article by two reviewers (MA and LD), who subsequently met to cross-check extracted information against the original articles.

2.3. Data extraction and synthesis

The full texts of the remaining studies were retrieved and analyzed using an extraction form. We grouped the studies in four categories: LESS variants, reference scores, differences according to sex and differences according to sport.

The data extraction process included capturing the following information: authors, year of publication, sample size, population, age of participants, results, and conclusions. In the case of articles focusing on LESS variants, the information extracted included the variant name and the characteristics of the observed changes. For articles related to normal values and differences between sexes or sports, the extracted data included LESS score and specific variations observed in terms of plane of landing or item of the test changed. Included records underwent qualitative and quantitative analysis via narrative synthesis.

3. Results

3.1. Characteristics of studies

The strategy used to search, evaluate and include studies according to PRISMA-ScR guidelines is represented in Fig. 1.

The 20 studies included in the analysis involved a total of 18,093 participants (age = 8–30 years, males = 70.6%). The military population was the most frequently analyzed, with a total of 7 studies (n = 16,603).

Six studies used some variant of the LESS, 5 studies described normal or reference scores, 3 studies investigated differences according to sex, and 6 studies examined differences based on the sport practiced. One study focused on normal values and sex differences.

3.2. LESS variants

In addition to the original version of the LESS, several modified versions were also used. The Single-Leg Landing Error Scoring System

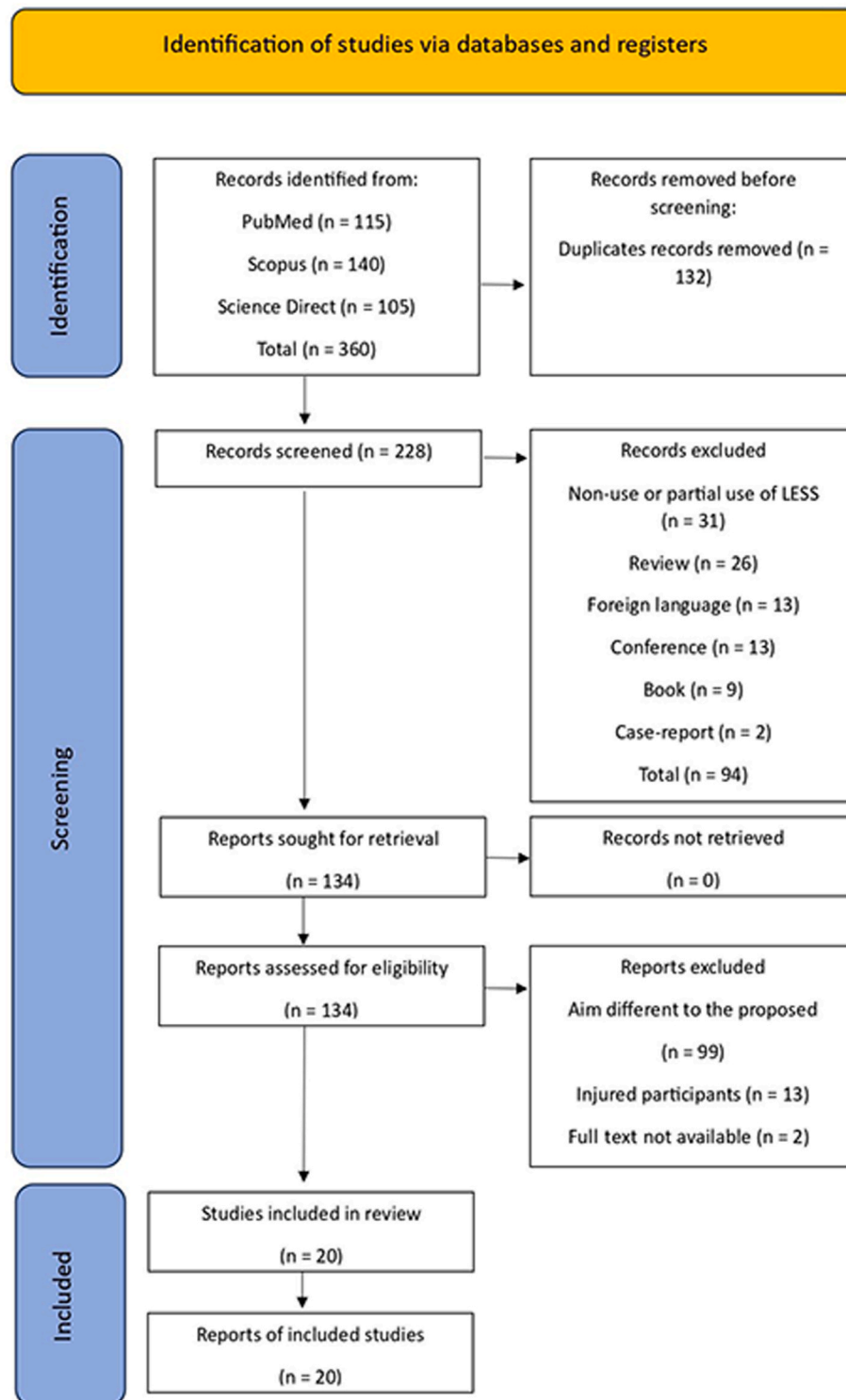


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta- Analyses (PRISMA) flow diagram of the screening process.

(SL-LESS), described by O'Connor (O'and Connor, 2015) proposes the participant standing on a 20 cm box on a single leg and then jumping out a distance of 25% of their height and landing with that same leg. Using this version Mifsud et al. found differences in the loading patterns of the dominant leg between safe and high-risk landing techniques among state-level athletes, but they included only 11 items (Mifsud et al., 2022). Another variation involving jumping on a single leg was proposed by Measson et al. The authors combined the LESS and the Simple Hop Distance (SHD) test, termed the Single Hop Distance Landing Error Scoring System (SHD-LESS). Items 1, 2, 3, 8, and 9 were modified to suit the SHD task. A good intra-rater reliability was observed for the overall SHD-LESS scale scores among non-professional athletes engaged in cutting/pivoting sports (Measson et al., 2022).

Three studies included a military sample. One of them found good inter-rater reliability in a real-time version of the LESS (LESS-RT). Four trials were scored in real time including 10 jump-landing characteristics by raters with great experience in the method (Padua et al., 2011). The same version did not find predictive value for knee overuse, meniscal, and anterior cruciate ligament injuries (Schwartz et al., 2020). LESS-RT was also used in recreational athletes (Oğul & Ercan, 2022). The other study added 5 error items (LESS-22) to establish normative scores within the population (De la Motte et al., 2016). The LESS-22 has been developed following Cameron et al. (Cameron et al., 2022) data collection. It includes 5 additional test items: asymmetrical loading, asymmetrical heel-toe or toe-heel landing, excessive trunk-flexion displacement, asymmetrical timing, and real-time "wobble" (Eckard et al., 2022). Some authors have ventured into the automatic analysis of the evaluation of the record through monitoring with motion capture marker systems, using LESS-22 (Mauntel et al., 2017) or LESS-17 (Dar et al., 2019; Taborri et al., 2021) criteria.

Finally, Hanzlíková & Hébert-Losier assessed the impact of landing distance on the Landing Error Scoring System (LESS) among young active individuals. Participants executed jumps using self-selected distance (DSS) and the test's original distance (50% of body height), revealing that employing DSS could result in distinct scores and risk

categorization (Hanzlíková et al., 2021b).

All variants were conducted in young adults (aged between 15 and 25 years old). Table 1 summarizes the information about LESS variants.

3.3. Reference scores

Five (5) studies focused on finding normative scores for different populations. Two of them included military participants: the first one, with a large sample size ($n = 7591$), analyzed 12 cohorts of cadets and found an overall score of 4.99 ± 2.15 , with most of the errors in the sagittal plane (Eckard et al., 2021). The second one, with a smaller sample size ($n = 78$) found an overall value of 7.1 ± 2.1 with the 68% of participants with poor landing (Ericksen et al., 2021). Similar values were obtained on 36 elite rugby players (overall 7.01 ± 2.38 and 58% of participants with poor landing)(Rowell & Relph, 2021). Smaller values (2.56 ± 1.25 for males and 2.81 ± 1.64 for females) were informed for professional soccer players ($n = 67$) whose ages range from 16 to 31³⁶. Data from 233 youth volleyball players from 10 to 13 years old was obtained by Limroongreungrat et al. While the values were not informed, the authors show the proportion of athletes in each landing category: 30% had excellent (≤ 4), 16.7% had good (>4 to ≤ 5), 25.3% had moderate (>5 to ≤ 6) and 27.9% had poor (>6) values (Limroongreungrat et al., 2022). All the test were performed with LESS-17.

Except for the study conducted in volleyball players, all studies had a majority (or only) of male participants. This should be taken into account in the light of the existing differences between sexes, which is explained below. Table 2 summarizes the information about reference scores.

3.4. Differences according to sex

The articles included in this section expressed explicitly, in their objectives or hypothesis, the aim of analyzing differences in the LESS values according to sex. Four studies were included, with nearly 3.000 individuals analyzed in total, with ages close to 20 years old, both in

Table 1
– LESS variants.

Study (year)	Sample size (n)	Population	Age (years) Mean $\pm$ SD. Range	Variant name/LESS Score	Result/Conclusion
Mifsud et al. (2022)	n: 31 M: 10 F: 21	State-level athletes	Overall: 20.1 ± 4.1 M: NS F: NS LR: 20.7 ± 5.0 HR: 19.9 ± 3.9^c	Single-Leg Landing Error Scoring System (SL-LESS)	Safe and high-risk landing techniques expose athletes to significantly different loading patterns during double and single-leg dominant movements, but not in the non-dominant leg
Measson et al. (2022)	n: 19 M: 14 F: 5	Non-professional athletes from a cutting/pivoting sport	Overall: 22.4 ± 0.3 M: NS F: NS	Single Hop Distance Landing Error Scoring System (SHD-LESS)	The intra-rater reliability was good (ICCdom: 0.77; ICCnondom: 0.87; ICCpooled: 0.87) for the overall SHD-LESS scale scores
Padua et al. (2011)	n: 43 M: 19 F: 24	Freshmen at the Military Academy	Range: 18–23. M: NS F: NS	Real-time version of the LESS (LESS-RT)	Interrater reliability ranged from 0.72 to 0.81
De la Motte et al. (2016)	n: 1714 M: 1434 F: 280	Military applicants	Overall: NS M: 20.8 ± 3.0 F: 20.9 ± 3.2^c	LESS with 22 items (LESS-22)	Normative overall means was 5.7 ± 2.1 . For males was 5.5 ± 2.1 and for females was 6.5 ± 1.8
Hanzlíková et al. (2021b)	n: 70 M: 34 F: 36	Young active individuals	Overall: NS M: 18.9 ± 0.8 F: 19.6 ± 2.4^c	LESS with self-selected distance (SSD-LESS).	Using self-selected distance during the LESS might lead to different LESS errors and risk categorizations at an individual level than using 50% of body height
Schwartz et al. (2020)	n: 2474 M: 1663 F: 811	Soldiers	Overall: 19.4 ± 2.3 M: NS F: NS	Real-time version of the LESS (LESS-RT)	The LESS-RT has no predictive value for knee overuse injuries, meniscal injuries, and anterior cruciate ligament injuries

NS: Not Specified, LR: Low risk, HR: High risk.

a: Significantly different ($p < 0.05$).

b: No significant difference.

^cNot mentioned significance/Does not apply.

Table 2

Reference scores in healthy individuals.

Study (year)	Sample size (n)	Population	Age (years) Mean $\pm$ SD. Range	Scores	Result/Conclusion
Eckard et al. (2021)	n: 7591 M: 5666 F: 1925	Military Academy cadets	Overall: NS M: NS F: NS	Overall: 4.99 ± 2.15 Sagittal plane: 2.13 ± 1.40 Frontal/transversal: 1.67 ± 1.20 Plane-independent: 1.19 ± 0.55^b	Lower extremity movement quality assessed on the LESS was stable in incoming USMA cadets during the years 2005–2017
Bernardes Marques et al. (2022)	n: 67 M: 36 F: 31	Professional soccer players	Range: 16–31 M: 19.8 ± 1.4 F: 20.8 ± 1.6^b	Overall: M: 2.56 ± 1.25 F: 2.81 ± 1.64^b	No significant difference between groups was found for players performance on LESS
Rowell and Relph (2021)	n: 36	Male elite rugby players	Overall: 24.9 ± 4.63	Overall: 7.01 ± 2.38 LESS categories Excellent (≤ 4): 3% Good (4–5): 17% Moderate (5–6): 22% Poor (> 6): 58% ^b	The majority of players presented moderate to poor LESS scores, indicating an area of concern in this population
Limroongreunrat et al. (2022)	n: 233 M: 97 F: 136	Youth volleyball players	Overall: 12.0 ± 1.0 M: 12.2 ± 0.9 F: 11.9 ± 1.0^b	Overall: NS LESS categories Excellent (≤ 4): 30% Good (> 4 to ≤ 5): 16% Moderate (> 5 to ≤ 6): 25.3 % Poor (> 6): 27.9% ^b	The data may be applied to identify youth volleyball players at potential risk of sustaining a lower body injury from poor landing biomechanics
Ericksen et al. (2021)	n: 78 M: 49 F: 29	Military	Overall: 19.7 ± 2.3 M: NS F: NS	Overall: 7.1 ± 2.1 LESS categories Excellent (≤ 4): 6.4% Good (> 4 to ≤ 5): 12.8% Moderate (> 5 to ≤ 6): 12.8% Poor (> 6): 68% ^b	Army Reserve Officers' Training Corps population demonstrated poor jump landing mechanics

NS: Not Specified.

a: Significantly different ($p < 0.05$).

aNo significant difference.

bNot mentioned significance/Does not apply.

males and females.

The average values ranged from 2.5 to 6.2 for males, and from 2.8 to 7.6 for women. Two of them (Beutler et al., 2009; Oğul & Ercan, 2022) found higher values in females than in males (including the one with the largest sample, $n = 2753$), and the others did not find any significant differences (Lam et al., 2014; Marques et al., 2022). Besides the values, differences according to the plane or the item of LESS was also mentioned. Males demonstrated more at-risk landing movement patterns on the sagittal plane whereas females demonstrated more at-risk landing movement patterns on the frontal plane (Lam et al., 2014). Most of the studies agree that females tend to land and descend with medial knee position and displacement (valgus) (Beutler et al., 2009; Lam et al., 2014; Marques et al., 2022). Males were more likely to land with the heel first (Beutler et al., 2009; Lam et al., 2014), with an asymmetric foot landing (Beutler et al., 2009), a narrow stance width and the foot in external rotation (Marques et al., 2022). Table 3 summarizes the information about differences according to sex.

3.5. Differences according to sport

The articles included in this section expressed explicitly, in their objectives or hypothesis, the aim of analyzing differences in the LESS values according to type of sport, level of competition and/or number of sports that are participating now or that participated in the past.

The average values ranged from 3.2 for football players to 8.4 for soccer players. In male adolescents and adults, the level of competition and sports specialization did not influence the LESS values (Beese et al., 2015; Theiss et al., 2014; Venrick et al., 2021). In younger boys and girls (8–14 years), multisport athletes were 2.5 times more likely to have

fewer than 5 landing errors than the single-sport group (DiStefano et al., 2018). Females reporting moderate sport specialization had improved movement quality and significantly better LESS scores, compared to those with high/low specialization (Venrick et al., 2021). In recreational athletes, previous participation in multiple high school varsity sports that feature significant landing and cutting athletic demands (basketball, lacrosse, soccer, or volleyball) was associated with lower LESS scores compared to those who had a history of single-sport or no participation in these sports at this level (Herman et al., 2019).

Finally, between female athletes, soccer players seem to have better LESS values than basketball and volleyball players, but they were significantly younger too (Harato et al., 2022). Table 4 summarizes the information about differences according to sport.

4. Discussion

The aim of this study was to explore and summarize current literature about LESS variants, reference scores and differences according to sex and sport.

Regarding LESS variants, the SL-LESS consisted in a modified jump on one leg, unlike the traditional LESS which evaluates a jump with both legs. There are some biomechanical differences between these jumps. The single leg jump increases the muscle activation (mainly gastrocnemius and vastus medialis) and the joint stiffness (Tai et al., 2018; Van Soest et al., 1985; Wang & Peng, 2014). The different loading pattern between high and low risk landing in the SL-LESS is in agreement with previous studies (Bates et al., 2019; Hewett et al., 2005). A good inter-rater reliability for a real time version of LESS was also described. This could reduce the analysis time, as well as automatic software

Table 3
Differences according to sex.

Study (year)	Sample size (n)	Population	Age (years) Mean $\pm$ SD. Range	Scores	Result/ Conclusion
Lam & McLeod (2014)	n: 215 M: 116 F: 99	Intercollegiate athletes	Overall: NS M: 19.4 $\pm$ 1.5 F: 19.1 $\pm$ 1.1 ^b	Overall: NS M: 5.1 $\pm$ 2.5 F: 5.8 $\pm$ 2.3 ^b	Female athletes tend to demonstrate more improper movements on the frontal plane compared with male athletes
Beutler et al. (2009)	n: 2753 M: 1707 F: 1046	Military Academy cadets	Range: 18–24 M: NS F: NS	Overall: NE M: 4.65 $\pm$ 1.69 F: 5.34 $\pm$ 1.51 ^a	Female had worse LESS score. Qualitative movement scores were analyzed using factor analyses, yielding five factors, or “patterns”, contributing to poor landing technique
Bernardes Marques et al. (2022)	n: 67 M: 36 F: 31	Professional soccer players	Range: 16–31 M: 19.8 $\pm$ 1.4 F: 20.8 $\pm$ 1.6 ^c	Overall: NS M: 2.56 $\pm$ 1.25 F: 2.81 $\pm$ 1.64 ^b	No significant difference between groups was found for players performance on LESS
Ogul & Ercan (2022)	n: 55 M: 27 F: 28	Recreational athletes	Overall: 21.8 $\pm$ 1.1 M: 21.8 $\pm$ 1.2 F: 21.7 $\pm$ 0.9 ^b	Overall: 6.93 $\pm$ 1.99 M: 6.2 $\pm$ 2.0 F: 7.6 $\pm$ 1.8 ^a	Females scored significantly higher than males

NS: Not Specified.

^bNo significant difference.

^cNot mentioned significance/Does not apply.

^aSignificantly different ($p < 0.05$).

analysis. Finally, a LESS variant allowed the participants to choose the distance of jump, instead of using the 50% of their heights. The modification of the distance changes the shear force of the joint in a jump (Simpson & Pettit, 1997) and probably can't be interchangeable with the original test.

In terms of normative scores, these were identified for military and three cutting sports: rugby, soccer and volleyball. The total number of participants evaluated is relatively small, especially referring to athletes, compared with the same information available for other similar tests, like functional movement screen (FMS)(Pollen et al., 2021; O'Brien et al., 2022). The overall values reported were “moderate to poor” (i.e. values higher than 5 points), with small portions of the sample obtaining excellent to good results. The worst results were found in the military population and in volleyball players. In fact, for both populations, a high percentage of participants obtained scores considered risky. We believe it important to use LESS to observe individual errors, not only in those participants who obtained risky results but even in athletes with low

risk, for personalized training. Having reliable information regarding normative scores is important because it allows comparison with other populations, players or teams. It also improves risk stratification, which is crucial for planning and organization in a prevention program, mainly in cutting sports, because the change of direction, particularly side-step maneuvers, has been associated with non-contact ACL injuries in numerous sports (Brophy et al., 2015; Montgomery et al., 2018; Waldén et al., 2015a). The present review has revealed the absence of normative scores in the literature on sports closely associated with this injury, like football, basketball or handball, representing an important gap in the available knowledge.

With respect to differences in LESS score according to sex, the results showed that females seem to have worse LESS scores than males, but no meta-analysis was conducted, and most studies included male participants. There are some possible explanations for a lower performance in females landing. Decreased potential for dynamic stabilization of the knee joint and uncoordinated recruitment patterns of the lower extremity muscles has been reported previously in females (Henry & Kaeding, 2001). Additionally, in adolescence, girls do not experience a rapid increase in muscle strength like boys do and it seems that the development of their hamstring muscle strength falls behind that of their quadriceps (Wild et al., 2012). Females also have more landing errors related to valgus, an anatomical characteristic which increases with age, especially during adolescence (Holden et al., 2016; Pappas & Carpes, 2012). However, it's probably that sex differences in athletic performance tends to diminish or disappear with increasing skill (Bruton et al., 2013). Males have more errors related to landing with the heels, a recognized factor for ACL injury (Koga et al., 2018; Montgomery et al., 2018; Waldén et al., 2015b). Consistent with this, Marquez et al. found different neuromuscular control in males and females, with greater EMG activity in the tibialis anterior of males during landing (Márquez et al., 2017).

Finally, regarding sports, results showed that sport specialization, mainly in its early stage, has been associated with increased injury risk (Jayanthi et al., 2015; Pasulka et al., 2017; Post et al., 2017). The influence of sports specialization in LESS score was not clear. Some research investigated the effect of specialization during landing tasks. Youth competitive gymnasts have differences in some balance tests according to the level of specialization, but this isn't associated with the errors in landing (Root et al., 2019). The study conducted by Di Cesare et al. revealed biomechanical alterations that suggest a potential compromise in neuromuscular control among sport-specialized female athletes (DiCesare et al., 2019). Soccer players had better LESS values in a sample that also included basketball and volleyball players, despite both sports involving more jumps than football, but is in line with Bernardes Marques et al. who reported low values for soccer players too (Marques et al., 2022).

4.1. Recommendations for future research

Most of the studies included young adults (in their twenties or younger). In fact, the oldest age included was 31 in a cohort of professional football players. The absence of data in older populations should be particularly noted given the increase of injury risk with aging (Green & Pizzari, 2017; Opar et al., 2012; Stevenson et al., 2000). A similar situation occurs with sex. More than 70% of the total population assessed in the included studies was male. Since the LESS has potential as a screening tool to determine ACL injury risk (Padua et al., 2015) and females are more likely to suffer this injury (Acevedo et al., 2014; Bisciotti et al., 2019; Montalvo et al., 2019), more research in this population is encouraged. Given the scarce number of studies available, more research should be done before conducting a systematic review and meta-analysis. More normative scores are needed, especially about athletes. Reference values in basketball, hockey, and many other popular sports are still not available. Samples with female participants and participants older than 30 years old should be included in further

Table 4

– Differences according to sport.

Study (year)	Sample size (n)	Population	Age (years) Mean ± SD. Range	Scores	Result/Conclusion
Harato et al. (2022)	n: 42 Basketball: 25 Soccer: 8 Volleyball: 9	Female athletes	Overall: NS Basketball: 20.4 ±1.4 Volleyball: 20.6 ±1.1 Football: 18.9 ±1.0 ^a	Overall: NS Basketball: 4.8 ± 1.5 Volleyball: 6.0 ± 2.8 Football: 3.3 ± 1.4 ^a	Female basketball and volleyball players have worse LESS, smaller peak knee flexion angle, greater knee abduction and knee internal rotation at IC, compared to female soccer players
Venrick et al. (2021)	n: 1950 M: 1491 F: 459 Sports specialization LL: 1045 ML: 600 HL: 305	Military Academy cadets	Range: 17–23 M: NS F: NS	Sports specialization LL: 4.48 ± 1.96 ML: 4.28 ± 1.94 HL: 4.44 ± 1.89 ^b	Female reporting moderate sport specialization had improved movement quality and significantly better LESS scores compared to those with high/low specialization
Theiss et al. (2014)	n: 277 M: 222 F: 55 Level of competition IM: 152 CC: 59 DI: 66	Military Academy cadets	Overall: 19.3 ± 0.8 M: NS F: NS IM: 19.3 ± 0.9 CC: 19.4 ± 0.7 DI: 19.2 ± 0.8 ^b	Overall: NS IM: 5.39 ± 0.29 CC: 5.26 ± 0.27 DI: 5.06 ± 0.28 ^b	No significant differences in landing errors between the levels of athletic group
Beese et al. (2015)	n: 40 Sports specialization SS: 21 MS: 19	Female athletes	Overall: NS SS: 15.1 ± 1.2 MS: 15.3 ± 1.2 ^c	Overall: 6.48 ± 1.89 SS: 6.84 ± 1.81 MS: 6.07 ± 1.93 ^b	Participation in soccer alone compared with participation in multiple sports did not affect LESS scores in adolescent female soccer players
Herman et al. (2019)	n: 50 M: 28 F: 22 Previous varsity sports participation OVS: 11 1VS: 21 2VSs: 18	Recreational athletes	Overall: 23.8 ± 2.5 OVS: 23.4 ± 3.1 1VS: 23.8 ± 2.5 2VSs: 24.1 ± 2.2 ^b	Overall: NS OVS: 5.89 ± 1.2 1VS: 5.38 ± 1.93 2VSs: 3.56 ± 1.97 ^a	Previous participation in multiple high school VSs that feature significant landing and cutting athletic demands was associated with lower LESS scores in adult recreational athletes
DiStefano et al. (2018)	n: 355 M: 122 F: 233 Sports specialization SS: 91 MS: 264	Youth athletes	Range: 8–14 years. M: NS F: NS SS: 11 ± 2 MS: 11 ± 2 ^b	Overall: NS SS: 6.42 ± 1.74 MS: 6.06 ± 1.84 ^b	Sport sampling at a young age is associated with improved neuromuscular control, which may reduce injury risk in youth athletes

NS: Not Specified, LL: Low Level, ML: Moderate Level, HL: High Level, IM: Intramural, CC: Competitive Club, DI: NCAA Division I, SS: Single-sport, MS: Multi-sport, VS: Number of Varsity Sports.

^a Significantly different ($p < 0.05$).

^b No significant difference.

^c Not mentioned significance/Does not apply.

research.

4.2. Limitations and future work

Some limitations of the present review must be mentioned. Some included studies did not mention the scores obtained. Instead, they only informed the proportion of subjects at each risk categorization. We did not conduct a meta-analysis, so it is difficult to provide a quantitative summary or combine the results of the studies in a statistical manner. Also, as this is a scoping review with the aim of presenting the available research currently on the literature, it was not within its scope to evaluate the quality of the included studies. Finally, the heterogeneity of the studies can make it challenging to pool the results or draw definitive conclusions.

The protocol for this scoping review included only conventional subject searching in databases as the literature searching technique. The reasons behind this choice were: 1) the reported efficiency of this method in finding appropriate articles (Papaioannou et al., 2010), 2) the publications on this subject have well-identified keywords and the topics are not broad or ill defined, which allowed us to presume that the database search would reveal the appropriate studies (Heath et al.,

2022) and 3) the objective of a scoping review, which is to identify the nature and extent of research evidence to determine if a full systematic review is needed (Grant & Booth, 2009; Munn et al., 2022). In making this decision, the risk of missing some studies that meet the inclusion criteria was assumed, and a more comprehensive search strategy would be implemented if a systematic review is conducted.

5. Conclusion

The variations for the LESS included conducting the test with a single leg, modifying the jump distance, adding additional items, and scoring in real-time. Regarding the reference values, most of them were obtained for a military population. Males and females show different pattern landing with errors in different planes, which must be taken into account when evaluating athletes. Our findings support the need for more field studies to determine reference scores for the LESS, especially in females, participants older than 30 years old, and for basketball or hockey players. The automatization of the analysis represents an opportunity, but the reliability of the method must be evaluated. More research is needed before conducting a systematic review and meta-analysis.

Ethical approval

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Ethical statement

Ethical approval is not required due to the nature of the study.

CRediT authorship contribution statement

Maximiliano Ezequiel Arlettaz: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lucas Nahuel Dorsch:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paola Andrea Lucia Catalfamo-Formento:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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